Two Production Lines, Two Sampling Plans

AP Statistics · Unit 4 · Topic 4.1 · B: 2027-01-25 · E: 2027-01-27 · TEACHER KEY

CED TETHER

- **Enduring Understanding: UNC-3** — Probabilistic reasoning allows us to anticipate patterns in data.
- Skills:
- **Skill 3.B** — Determine parameters for probability distributions.
- **Skill 3.C** — Describe probability distributions.
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- Learning Objectives and Essential Knowledge:
- UNC-3.Q Determine parameters for a sampling distribution for sample means. [Skill 3.B]
- UNC-3.Q.1 For a numerical variable, when random sampling from a population with mean μ and standard deviation σ, the sampling distribution of the sample mean has mean $\mu\_\bar{x} = \mu$ and standard deviation $\sigma_\bar{x} = \sigma/\sqrt{}$n. UNC-3.Q.2 If sampling without replacement and sample size is less than 10% of the population size, the difference is negligible.
- UNC-3.R Determine whether a sampling distribution of a sample mean can be described as approximately normal. [Skill 3.C]
- UNC-3.R.1 For a numerical variable, if the population distribution can be modeled with a normal distribution, the sampling distribution of the sample mean can be modeled with a normal distribution. UNC-3.R.2 For a numerical variable, if the population distribution cannot be modeled with a normal distribution, the sampling distribution of the sample mean can be modeled approximately by a normal distribution provided the sample size is large enough (n $\geq$ 30).
- UNC-3.S Interpret probabilities and parameters for a sampling distribution for a sample mean. [Skill 4.B]
- UNC-3.S.1 Probabilities and parameters for a sampling distribution for a sample mean should be interpreted using appropriate units and within the context of a specific population.

Essential question: How do population shape, sample size, and sampling variability determine whether a normal probability for a sample mean is trustworthy?

DOK-3 skill: 4.B · **FRQ pattern:** `normal-vs-clt-model-adjudication`

DOK rationale: Part (c) adjudicates two incomplete normal-model claims by coordinating population shape, sample size, the 10 percent condition, sampling-distribution variability, and the consequence of rejecting a valid probability model.

First take → rules + (a)–(c) → turn in **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Ask students to mark one plausible clause in each claim without confirming either.
- At minute 5, read the rules box aloud once; students start (a).
- Return to the board slide at minute 33. Circulate and require a separate normality justification for each line.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Mark a plausible clause in each claim and cite population shape or sample size.

- Use the rules box to check each model before calculating its normal tail probability.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- Is 30 the only route to a normal sampling distribution?
- Does the Central Limit Theorem change the original population or the distribution of sample means?
- Why are both sampling distributions centered at 40 despite different sample sizes?
- How do standard deviations 4 and 2 change where 46 falls on each model?

Adult role

Solo teacher. Circulate 5 pairs. Ask students to evaluate each clause of each claim, then connect the normality path and sampling standard deviation to its tail probability.

[WARN] LOOK FOR

- Using n at least 30 as a universal requirement even for a normal population.
- Claiming a skewed population prevents the CLT from applying at n equals 36.
- Using the population standard deviation of 12 for both distributions of sample means.

SCORING PART (c) — E / P / I

Expected elements

- **judgment-1** (required) — Supports both approximations, using A population normality and B n=36 CLT route.
- **judgment-2** (required) — Explains the correct and incorrect portion of both students' claims.
- **judgment-3** (required) — Connects SDs 4 and 2, or z-scores 1.5 and 3, to the different tail probabilities.

E	Meets all three checks with correct contextual reasoning; equivalent wording is acceptable.
P	Shows at least one substantive correct check, but has an omission or error that prevents E.
I	Shows none of the three checks with substantive correct reasoning; a choice alone is insufficient.

[STAR] TODAY'S PROBLEM — annotated

Suppose Lines A and B each contain 1,000 cycle times with population mean 40 seconds and standard deviation 12 seconds. Line A is approximately normal and an SRS of $n_A = 9$ is selected; Line B is strongly right-skewed and an SRS of $n_B = 36$ is selected. An auditor wants $P(\bar{X} > 46)$ for each.

Serena: "Only B can use a normal model because $36 \geq 30$; its probability is 0.00135. A is too small."

Mateo: "Only A can use a normal model because its population is normal; its probability is 0.0668. B is too skewed."

Sampling plans

Line	Shape	N	n	μ	σ
A	Normal	1000	9	40	12
B	Right-skewed	1000	36	40	12

FIRST TAKE (5 min)

Which plan seems more likely to give averages close to 40 seconds? Give one reason from the sample sizes or shapes.
Key: Not graded. Each student is correct about one line and wrong about the other; the finish should replace either-or reasoning with the two separate normality pathways.

Rules callout "MODEL SAMPLE MEANS, NOT INDIVIDUAL VALUES (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** State the mean of each sampling distribution and the 10% condition for using $\sigma_{\bar{X}} = \sigma/\sqrt{n}$.

Key: Both sampling distributions have mean 40 seconds. Without replacement, each sample must be no more than 10 percent of its population to use $\sigma_{\bar{X}} = \sigma/\sqrt{n}$ without a finite-population correction.

DOK 2 **(b)** Verify the 10 percent condition, compute both sampling-distribution standard deviations, and calculate each normal tail probability when justified.

Key: Both SRS sizes satisfy the cap $0.10(1{,}000) = 100$. For A, $\sigma_{\bar{X}} = 12/\sqrt{9} = 4$; its normal population justifies $z = (46 - 40)/4 = 1.5$ and an upper tail of 0.0668. For B, $\sigma_{\bar{X}} = 12/\sqrt{36} = 2$; the CLT applies because $36 \geq 30$, giving $z = 3$ and an upper tail of 0.00135.

DOK 3 **(c)** In 3–4 sentences, explain what each student gets right and wrong. Use the two normality pathways to decide which probabilities are supported. Connect the different tail probabilities to the sampling SDs in (b).

Key: Neither whole claim is complete. Serena is right about B: despite population skew, $n_B = 36$ activates the CLT and supports 0.00135; she wrongly rejects A, whose normal population supports a normal sampling distribution at $n_A = 9$. Mateo is right about A and 0.0668 but ignores B's CLT route. Their rules would each discard one valid probability. Both models center at 40, but 46 is 1.5 sampling SDs above A's center and 3 above B's, so B's larger sample produces the smaller tail.

EXIT

One thing the rules box changed about my first take: ______________________________